Zero-Chatter Multi-Agent Code Synthesis via Arrow Flight State Embeddings, Isotropy-Guided Evolution, and Fail-Closed Integration Gates

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Abstract

Multi-agent large language model (LLM) systems for software engineering typically coordinate via natural-language message passing ("chatter"). As the number of concurrent agents K grows, conversational coordination suffers from three structural failure modes: (1) quadratic context-window inflation $\mathcal{O}(K^2 \cdot \bar{L})$ that crowds out repository evidence, (2) silent directional mode collapse in which parallel agents converge onto redundant or mutually incompatible patches, and (3) unverified branch integration that regresses baseline repository metrics. We present LEIO-HARNESS, a zero-chatter multi-agent execution and evolution runtime implemented in Rust. Instead of exchanging natural-language transcripts, concurrent agents execute in isolated Git worktrees and coordinate exclusively by streaming L_2-normalized state and intent embeddings over an Apache Arrow Flight binary bus (`do_put`, `do_get`, and bidirectional `do_exchange` cosine top-k matching). To explore diverse synthesis trajectories without gradient access, LEIO-HARNESS introduces an embedding-space evolutionary operator (EMBEDDINGGEPA) that combines geodesic spherical linear interpolation (`slerp`), tangent-space Gaussian perturbation bounded by a hard geodesic trust-region cap ($\theta_{\max}$), and semantic-anchor drift regularization across a multi-objective Pareto archive. To prevent population mode collapse prior to branch merge, we introduce a runtime SIGREG gate that computes the closed-form sliced Epps–Pulley characteristic-function statistic against $\mathcal{N}(0,1)$ over random unit projections $z = \sqrt{d}(Xu)$. Finally, a fail-closed integration gate merges candidate worktrees in a throwaway branch and promotes via fast-forward if and only if the integrated tree is conflict-free, satisfies the SIGREG isotropy bound ($\tau_{\text{iso}} = 2.0$), and strictly Pareto-dominates the single-agent baseline with a cryptographic SHA-256 evidence digest. Across 88 multi-model parallel lanes spanning 8 frontier and flash models, LEIO-HARNESS eliminates inter-agent token chatter (100% reduction in coordination tokens), cleanly separates collapsed populations ($I_{\text{EP}} = 3.504 > 2.0$) from isotropic exploration ($I_{\text{EP}} = 0.343 \leq 2.0$), and lifts small-model grounded evidence recall from 0.089 to 0.261 (+193%), surpassing larger frontier baselines at 6/6 schema validity.

1 Introduction

Autonomous software engineering increasingly relies on parallel ensembles of coding agents—explorers, workers, verifiers, and security reviewers—operating concurrently on a shared repository [1, 2, 3, 4]. In prevailing multi-agent frameworks, coordination is mediated through natural language: agents broadcast plans, critique intermediate diffs, and negotiate task boundaries inside a shared conversational transcript.

While intuitive, natural-language coordination imposes a severe *chatter tax*. First, when K concurrent agents exchange messages of mean length $\bar{L}$ tokens over R rounds, the cumulative coordination context scales as $\mathcal{O}(R \cdot K^2 \cdot \bar{L})$, rapidly exhausting attention capacity that should be reserved for repository source code, build logs, and formal specifications. Second, natural-language consensus biases parallel agents toward premature agreement—a phenomenon analogous to representation mode collapse in self-supervised learning [7, 10]—causing K parallel lanes to synthesize near-identical patches rather than exploring complementary regions of the design space. Third, existing multi-agent harnesses rarely enforce a formal improvement contract at integration time: merged branches from multiple agents can silently degrade secondary objectives (such as latency, binary size, or warning counts) relative to a single-agent baseline.

In this paper, we argue that multi-agent code synthesis should separate *intra-agent reasoning* (which uses structured repository context and compiler feedback inside an isolated worktree) from *inter-agent coordination* (which should operate over compact, fixed-width geometric representations with formal statistical guarantees). We design and implement LEIO-HARNESS, a production Rust runtime that replaces natural-language agent chatter with four tightly coupled mechanisms:

1. **Zero-Copy Arrow Flight Semantic Bus (§4.1):** Agents never read or write peer conversation transcripts. Instead, each lane publishes and queries L_2-normalized state and intent vectors $\mathbf{x} \in \mathbb{S}^{d-1}$ over Apache Arrow Flight (`do_put`, `do_get`, and bidirectional `do_exchange`), backed by an atomic, lock-guarded columnar snapshot store.

2. **Spherical Trust-Region Evolution (§4.2):** To diversify agent intents without leaving embedding space, EMBEDDINGGEPA evolves candidate vectors on the unit hypersphere $\mathbb{S}^{d-1}$ using geodesic crossover (`slerp`), tangent-projected Gaussian mutation clipped to a maximum angular trust region $\theta_{\max}$, and a semantic-anchor drift penalty $R_{\text{adj}} = R - \beta(1 - \langle \mathbf{x}, \mathbf{x}_{\text{seed}} \rangle)$ over a multi-objective Pareto archive.

3. **Sliced Epps–Pulley Isotropy Gate (§4.3):** Building on sliced characteristic-function tests for isotropic Gaussian representations (SIGREG) [9, 10], LEIO-HARNESS audits the population of lane embeddings before integration by projecting $\mathbf{X} \in \mathbb{R}^{K \times d}$ onto M deterministic unit directions $\mathbf{u}_m \in \mathbb{S}^{d-1}$ (seeded via `SplitMix64`) and testing $z = \sqrt{d}(\mathbf{X}\mathbf{u}_m)$ against $\mathcal{N}(0, 1)$.

4. **Fail-Closed Pareto Integration Gate (§4.4):** Candidate Git worktrees are merged into an ephemeral integration worktree and evaluated against the single-agent baseline ref. Fast-forward promotion into the target branch is permitted if and only if merge conflicts are zero, population isotropy satisfies $I_{\text{EP}}(\mathbf{X}) \leq \tau_{\text{iso}}$, no metric regresses beyond noise floor $\epsilon = 10^{-6}$, and at least one metric strictly improves, emitting a tamper-evident SHA-256 evidence digest.

2 Related Work

Conversational Multi-Agent Coding Frameworks. Systems such as ChatDev [1], MetaGPT [2], AutoGen [3], and Agentless [5] organize LLM agents into role-based pipelines or debate topologies. Empirical audits of multi-agent LLM systems [6] show that unconstrained natural-language message passing accounts for 60–85% of total token expenditure and frequently propagates hallucinations across agents. LEIO-HARNESS eliminates inter-agent natural-language transcripts entirely, restricting cross-lane communication to fixed-schema Arrow Flight vector frames and Git tree states.

Evolutionary Search for Prompts and Agent States. Evolutionary prompt optimization (e.g., EvoPrompt [11], Promptbreeder [12], and GEPA [13]) typically mutates discrete text strings

via LLM rewrites. In contrast, EMBEDDINGGEPA operates directly on continuous unit-hypersphere intent embeddings $\mathbb{S}^{d-1}$ using exact geodesic interpolation and tangent-space projection, ensuring deterministic, zero-LLM-cost state mutation bounded by an explicit angular trust region.

Isotropy and Mode Collapse in Representation Spaces. Representation collapse—where high-dimensional embeddings concentrate along a low-dimensional cone—has been studied extensively in contrastive and self-supervised learning [8, 7, 10]. We adapt the sliced Epps–Pulley omnibus test for univariate normality [9, 10] into a deterministic, dependency-free runtime gate that detects directional collapse across concurrent coding lanes before their Git branches are merged.

3 Problem Formulation

Consider a software repository $\mathcal{R}$ at baseline Git commit c_0 and a synthesis specification $\mathcal{T}$ (comprising a natural-language task goal, a structured repository context bundle $\mathcal{B}_\mathcal{T}$, and an executable objective evaluator $\mathcal{E}$). A multi-agent run spawns K concurrent lanes $\mathcal{L}_1, \ldots, \mathcal{L}_K$. Each lane $\mathcal{L}_k$ executes in an isolated Git worktree W_k branched from c_0, producing a candidate branch b_k and an intent/outcome embedding $\mathbf{x}_k \in \mathbb{S}^{d-1}$, where $\mathbb{S}^{d-1} = \{\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}\|_2 = 1\}$.

Running the objective evaluator $\mathcal{E}$ on any repository tree c yields a finite metric vector $\mathbf{m}(c) \in \mathbb{R}^P$, partitioned into maximization objectives $P_{\max}$ (e.g., test pass rate, grounded evidence recall) and minimization objectives $P_{\min}$ (e.g., latency, token cost). Define the signed improvement delta for objective $p \in \{1, \ldots, P\}$ between baseline c_0 and candidate integrated tree c_{int} as:

$$\Delta_p(c_0, c_{\text{int}}) = \begin{cases} m_p(c_{\text{int}}) - m_p(c_0), & \text{if } p \in P_{\max}, \\ m_p(c_0) - m_p(c_{\text{int}}), & \text{if } p \in P_{\min}. \end{cases} \tag{1}$$

Our goal is to synthesize an integrated commit $c_{\text{int}} = \text{Merge}(c_0, \{b_1, \ldots, b_K\})$ while:

1. Bounding inter-agent coordination bandwidth to $\mathcal{O}(K \cdot d)$ bytes independent of reasoning trace length;
2. Maintaining angular diversity across the population $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_K]^\top \in \mathbb{R}^{K \times d}$ while keeping each lane anchored near the user's seed intent $\mathbf{x}_{\text{seed}} \in \mathbb{S}^{d-1}$;
3. Promoting c_{int} to target ref c_{target} *if and only if* c_{int} is conflict-free, non-collapsed, and strictly Pareto-improves upon c_0.

4 Methodology: The leio-harness Architecture

4.1 Zero-Chatter Coordination via Apache Arrow Flight

To eliminate natural-language chatter between concurrent lanes, LEIO-HARNESS implements `SemanticBus`, an Apache Arrow Flight [14] gRPC server over a fixed 7-column schema:

$$\mathcal{S}_{\text{bus}} = (\texttt{seq} : \text{u64},\ \texttt{timestamp_ms} : \text{i64},\ \texttt{agent_id} : \text{utf8},\ \texttt{run_id} : \text{utf8},\ \texttt{topic} : \text{utf8},\ \texttt{dimension} : \text{u32},\ \texttt{vector} : \text{b} \tag{2}$$

where `vector` stores raw little-endian IEEE-754 `f32` bytes ($4d$ bytes per row, validated for $1 \leq d \leq 65{,}536$ and finite entries). The bus exposes three zero-copy Flight primitives:

- **`do_put` (Transactional Publish):** Lanes stream Arrow `RecordBatch` frames (1 to 4,096 rows per batch; 128 MiB bus capacity cap). Sequence numbers `seq` are monotonically assigned under an exclusive `fs2` file lock and atomically persisted to disk via write-temp-fsync-rename before memory commit.
- **`do_get` (Incremental Subscription):** Lanes pass a JSON ticket $\{\texttt{topic}, \texttt{since_seq}\}$ and receive a single `RecordBatch` containing all rows matching the optional topic filter with $\texttt{seq} > \texttt{since_seq}$.
- **`do_exchange` (Bidirectional Cosine Top-k Matching):** Lanes push a query embedding $\mathbf{q} \in \mathbb{R}^d$ in a single-row `RecordBatch` with descriptor $\texttt{cmd}$ = "match" and metadata (k, topic), and the server streams back the top-k most similar peer trajectories ranked by cosine similarity:

$$\text{sim}(\mathbf{q}, \mathbf{x}_i) = \frac{\langle \mathbf{q}, \mathbf{x}_i \rangle}{\|\mathbf{q}\|_2 \, \|\mathbf{x}_i\|_2}, \quad \text{sorted by } \text{sim}(\mathbf{q}, \mathbf{x}_i) \text{ descending, then } \texttt{seq}_i \text{ ascending.} \tag{3}$$

Additionally, `do_action` supports server-side embedding (`embed-and-put`) via a configurable local/remote embedding endpoint, falling back deterministically when offline.

4.2 Spherical Evolutionary State Search (EmbeddingGEPA)

When orchestrating evolutionary rounds across lanes, mutating text prompts via an LLM introduces nondeterminism, latency, and semantic drift. LEIO-HARNESS instead evolves intent embeddings directly on $\mathbb{S}^{d-1}$ using pure, deterministic Rust (`SplitMix64` bitstream + Box–Muller Gaussian transform).

Geodesic Crossover (`slerp`). Given two parent intent vectors $\mathbf{v}_0, \mathbf{v}_1 \in \mathbb{R}^d$, we normalize $\mathbf{a} = \mathbf{v}_0/\|\mathbf{v}_0\|_2$ and $\mathbf{b} = \mathbf{v}_1/\|\mathbf{v}_1\|_2$, compute the subtended angle $\omega = \arccos(\text{clamp}(\langle \mathbf{a}, \mathbf{b} \rangle, -1, 1))$, and interpolate along the great-circle arc for $t \in [0, 1]$:

$$\text{slerp}(\mathbf{a}, \mathbf{b}; t) = \text{normalize}\left(\begin{cases} (1-t)\mathbf{a} + t\mathbf{b}, & \text{if } |\sin\omega| < 10^{-6}, \\ \dfrac{\sin((1-t)\omega)}{\sin\omega}\mathbf{a} + \dfrac{\sin(t\omega)}{\sin\omega}\mathbf{b}, & \text{otherwise.} \end{cases} \right) \tag{4}$$

The linear fallback for $|\sin\omega| < 10^{-6}$ guarantees numerical stability in both parallel ($\omega \to 0$) and antipodal ($\omega \to \pi$) regimes.

Tangent-Space Trust-Region Mutation. Given a normalized parent $\mathbf{b} = \mathbf{p}/\|\mathbf{p}\|_2 \in \mathbb{S}^{d-1}$, mutation strength $\sigma > 0$, and maximum angular radius $\theta_{\max} > 0$ (in degrees), we sample an isotropic Gaussian vector $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ via Box–Muller and project it onto the tangent hyperplane $T_{\mathbf{b}}\mathbb{S}^{d-1}$ by stripping its radial component:

$$\boldsymbol{\eta}_\perp = \boldsymbol{\eta} - \langle \boldsymbol{\eta}, \mathbf{b} \rangle \mathbf{b}, \qquad \hat{\boldsymbol{\eta}}_\perp = \frac{\boldsymbol{\eta}_\perp}{\|\boldsymbol{\eta}_\perp\|_2}. \tag{5}$$

We form the unconstrained step $\tilde{\mathbf{c}} = \text{normalize}(\mathbf{b} + \sigma\hat{\boldsymbol{\eta}}_\perp)$ and compute its angular deviation $\theta = \frac{180}{\pi}\arccos(\text{clamp}(\langle \mathbf{b}, \tilde{\mathbf{c}} \rangle, -1, 1))$. If $\theta > \theta_{\max}$, we project $\tilde{\mathbf{c}}$ back onto the geodesic boundary of the spherical cap via $\text{slerp}(\mathbf{b}, \tilde{\mathbf{c}}; \theta_{\max}/\theta)$.

Semantic-Anchor Regularization and Pareto Archive. To prevent mutated intents from drifting away from the user's original specification $\mathbf{x}_{\text{seed}} \in \mathbb{S}^{d-1}$, each candidate's scalarized reward R is regularized by a cosine anchor penalty with weight $\beta \geq 0$:

$$R_{\text{adj}}(\mathbf{x}) = R(\mathbf{x}) - \beta(1 - \langle \hat{\mathbf{x}}, \hat{\mathbf{x}}_{\text{seed}} \rangle). \tag{6}$$

Candidates are maintained in a `ParetoFrontier` archive where candidate A dominates B ($A \succ B$) iff $\forall p,\ s_p(A) \geq s_p(B)$ and $\exists p,\ s_p(A) > s_p(B)$. Parent selection alternates between select_least_explored (minimizing evaluation count n_{eval}) and select_best_aggregate.

4.3 Sliced Epps–Pulley Isotropy Collapse Detection (SIGReg)

Even with distinct seeds or models, concurrent lanes can collapse into a narrow cone of the embedding space when guided by similar repository priors. Following the theoretical foundation of SIGReg [10] and the Epps–Pulley omnibus normality test [9], if K embeddings $\mathbf{X} \in \mathbb{R}^{K \times d}$ are distributed uniformly (isotropically) on $\mathbb{S}^{d-1}$, then for any fixed unit direction $\mathbf{u} \in \mathbb{S}^{d-1}$ the scaled 1-D projection $z_i = \sqrt{d}\langle \hat{\mathbf{x}}_i, \mathbf{u} \rangle$ converges in distribution to the standard normal $\mathcal{N}(0,1)$ as $d \to \infty$.

For each slice $m \in \{1, \ldots, M\}$, we draw $\mathbf{u}_m \in \mathbb{S}^{d-1}$ deterministically via `SplitMix64`, compute $z_i^{(m)} = \sqrt{d}\langle \hat{\mathbf{x}}_i, \mathbf{u}_m \rangle$ for $i \in \{1, \ldots, K\}$, and evaluate the closed-form Epps–Pulley statistic measuring the weighted L^2 distance between the empirical characteristic function $\hat{\phi}(t) = \frac{1}{K}\sum_{i=1}^{K} e^{\mathrm{i}tz_i}$ and the standard normal characteristic function $\phi_0(t) = e^{-t^2/2}$ under Gaussian weight $\omega(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$:

$$\begin{aligned} T_{\text{EP}}(\mathbf{z}) &= K \int_{-\infty}^{\infty} \left|\hat{\phi}(t) - \phi_0(t)\right|^2 \omega(t)\, dt \\ &= \frac{1}{K} \sum_{i=1}^{K} \sum_{j=1}^{K} \exp\left(-\frac{(z_i - z_j)^2}{2}\right) - \sqrt{2} \sum_{i=1}^{K} \exp\left(-\frac{z_i^2}{4}\right) + \frac{K}{\sqrt{3}}. \end{aligned} \tag{7}$$

Averaging across M slices yields the population isotropy score:

$$I_{\text{EP}}(\mathbf{X}; M) = \frac{1}{M} \sum_{m=1}^{M} T_{\text{EP}}(\sqrt{d}\, \hat{\mathbf{X}} \mathbf{u}_m). \tag{8}$$

Well-dispersed isotropic populations satisfy $I_{\text{EP}}(\mathbf{X}) \lesssim 0.57$, whereas directionally collapsed populations satisfy $I_{\text{EP}}(\mathbf{X}) \geq 2.7$. We calibrate the fail-closed decision boundary at $\tau_{\text{iso}} = 2.0$.

4.4 Fail-Closed Integration and Promotion Gate

Algorithm 1 formalizes the LEIO-HARNESS integration gate (`run_gate`). Crucially, the gate never mutates the target branch in place during evaluation: it creates an ephemeral worktree W_{int}, attempts an octopus/sequential merge of $\{b_1, \ldots, b_K\}$, runs the objective command $\mathcal{E}$ on both c_0 and W_{int}, evaluates the four-state verdict $\{$`Improved`, `NoImprovement`, `Regressed`, `Collapsed`$\}$, and guarantees worktree cleanup via `discard_worktree`.

5 Theoretical Guarantees

Proposition 1 (Geodesic Trust-Region Invariance)**.** *For any non-zero parent vector $\mathbf{p} \in \mathbb{R}^d$ ($d \geq 2$), mutation strength $\sigma > 0$, and trust angle $\theta_{\max} \in (0°, 180°)$, let $\mathbf{c} =$ `trust_region_mutate`$(\mathbf{p}, \sigma, \theta_{\max}, rng)$. Then $\|\mathbf{c}\|_2 = 1$ and the geodesic angle $\angle(\mathbf{p}, \mathbf{c}) = \frac{180}{\pi} \arccos\langle \frac{\mathbf{p}}{\|\mathbf{p}\|_2}, \mathbf{c} \rangle$ satisfies $\angle(\mathbf{p}, \mathbf{c}) \leq \theta_{\max}$ up to IEEE-754 `f32` rounding (10^{-4} deg).*

Algorithm 1 Fail-Closed Multi-Agent Integration and Promotion Gate (`run_gate`)

Require: Repository $\mathcal{R}$, baseline ref c_0, target ref c_{target}, lane branches $\{b_1, \ldots, b_K\}$, objective spec $\mathcal{E}$, isotropy threshold $\tau_{\text{iso}} = 2.0$, noise floor $\epsilon = 10^{-6}$

Ensure: Gate report $(\text{verdict}, \mathbf{m}(c_0), \mathbf{m}(c_{\text{int}}), \boldsymbol{\Delta}, I_{\text{EP}}, \text{sha256}, \text{promoted})$

$\mathbf{m}(c_0) \leftarrow \text{RunObjectiveOnRef}(\mathcal{R}, c_0, \mathcal{E})$
$(W_{\text{int}}, b_{\text{int}}, \text{merged}) \leftarrow \text{IntegrateLanes}(\mathcal{R}, c_{\text{target}}, \{b_1, \ldots, b_K\})$
if ¬merged **then** ▷ Merge conflict: fail-closed immediately
  $\text{DiscardWorktree}(\mathcal{R}, W_{\text{int}}, b_{\text{int}})$
  return $(\texttt{NoImprovement}, \mathbf{m}(c_0), \varnothing, \varnothing, \varnothing, \text{false})$
end if
$(\mathbf{m}(c_{\text{int}}), I_{\text{EP}}) \leftarrow \text{RunObjectiveOnWorktree}(W_{\text{int}}, \mathcal{E})$
for all $p \in P_{\max} \cup P_{\min}$ **do**
  $\Delta_p \leftarrow (m_p(c_{\text{int}}) - m_p(c_0))$ if $p \in P_{\max}$ else $(m_p(c_0) - m_p(c_{\text{int}}))$
end for
if I_{EP} is present and $I_{\text{EP}} > \tau_{\text{iso}}$ **then**
  verdict $\leftarrow$ `Collapsed`
else if $\exists p : \Delta_p < -\epsilon$ **then**
  verdict $\leftarrow$ `Regressed`
else if $\forall p : \Delta_p \leq \epsilon$ **then**
  verdict $\leftarrow$ `NoImprovement`
else
  verdict $\leftarrow$ `Improved`
end if
$h \leftarrow \text{SHA256}\big(\text{CanonicalJSON}(\mathbf{m}(c_0), \mathbf{m}(c_{\text{int}}), \tau_{\text{iso}}) \parallel \texttt{0x00}\big)$
promoted $\leftarrow$ false
if verdict == `Improved` $\wedge$ promote_enabled **then**
  promoted $\leftarrow \text{FastForwardRef}(\mathcal{R}, b_{\text{int}} \to c_{\text{target}})$
end if
$\text{DiscardWorktree}(\mathcal{R}, W_{\text{int}}, b_{\text{int}})$
return $(\text{verdict}, \mathbf{m}(c_0), \mathbf{m}(c_{\text{int}}), \boldsymbol{\Delta}, I_{\text{EP}}, h, \text{promoted})$

Proof. Let $\mathbf{b} = \mathbf{p}/\|\mathbf{p}\|_2 \in \mathbb{S}^{d-1}$. By Eq. (5), the tangent perturbation $\boldsymbol{\eta}_\perp = \boldsymbol{\eta} - \langle \boldsymbol{\eta}, \mathbf{b} \rangle \mathbf{b}$ satisfies $\langle \boldsymbol{\eta}_\perp, \mathbf{b} \rangle = \langle \boldsymbol{\eta}, \mathbf{b} \rangle - \langle \boldsymbol{\eta}, \mathbf{b} \rangle \|\mathbf{b}\|_2^2 = 0$. The normalized pre-clip child $\tilde{\mathbf{c}} = (\mathbf{b} + \sigma \hat{\boldsymbol{\eta}}_\perp)/\sqrt{1+\sigma^2}$ has unit norm and subtends angle $\theta = \frac{180}{\pi} \arctan(\sigma)$. If $\theta \leq \theta_{\max}$, $\mathbf{c} = \tilde{\mathbf{c}}$ satisfies the bound immediately. If $\theta > \theta_{\max}$, $\texttt{slerp}(\mathbf{b}, \tilde{\mathbf{c}}; t)$ with $t = \theta_{\max}/\theta \in (0,1)$ parametrizes the constant-speed geodesic on $\mathbb{S}^{d-1}$ from $\mathbf{b}$ ($t = 0$) to $\tilde{\mathbf{c}}$ ($t = 1$), so $\angle(\mathbf{b}, \text{slerp}(\mathbf{b}, \tilde{\mathbf{c}}; t)) = t \cdot \theta = (\theta_{\max}/\theta)\theta = \theta_{\max}$, and final normalization preserves $\|\mathbf{c}\|_2 = 1$. □

Proposition 2 (Sliced Epps–Pulley Collapse Separation)**.** *Let $\mathbf{X}_{coll} = [\mathbf{v}, \ldots, \mathbf{v}]^\top \in \mathbb{R}^{K \times d}$ ($K \geq 3, \|\mathbf{v}\|_2 = 1$) be a directionally collapsed population where all K lanes emit identical normalized state vectors. Then for every projection slice $\mathbf{u}_m \in \mathbb{S}^{d-1}$ with $z_m = \sqrt{d}\langle \mathbf{v}, \mathbf{u}_m \rangle$, the sliced Epps–Pulley statistic satisfies:*

$$T_{EP}(\mathbf{z}^{(m)}) = K \left(1 - \sqrt{2} \exp\left(-\frac{z_m^2}{4}\right) + \frac{1}{\sqrt{3}}\right) \geq K \left(1 + \frac{1}{\sqrt{3}} - \sqrt{2}\right) \approx 0.1631K. \tag{9}$$

In expectation over $\mathbf{u}_m \sim$ Unif$(\mathbb{S}^{d-1})$ where $z_m \xrightarrow{d} \mathcal{N}(0,1)$, $\mathbb{E}[I_{EP}(\mathbf{X}_{coll})] = K(1 + \frac{1}{\sqrt{3}} - \sqrt{\frac{4}{3}}) \approx$

Table 1: Empirical verification of SIGReg sliced Epps–Pulley isotropy ($K = 8, d = 8, M = 64$) and EmbeddingGEPA trust-region mutation in `leio-harness v2.6.5`.

Subsystem	Regime / Test	Measured Value	Theoretical / Bound	Gate Status
SIGReg (§4.3)	Collapsed 8-lane population ($\mathbf{x}_1 = \cdots = \mathbf{x}_8$)	$I_{\text{EP}} = 3.5042$	$\mathbb{E}[I_{\text{EP}}] \approx 3.381$ (> 2.0)	`Collapsed` (Blocked)
	Isotropic 8-lane population ($\mathbf{x}_k \sim \text{Unif}(\mathbb{S}^7)$)	$I_{\text{EP}} = 0.3432$	$\mathbb{E}[I_{\text{EP}}] \approx 0.423$ (≤ 2.0)	`Healthy` (Passed)
	Collapse-to-Isotropic Separation Ratio	$10.21\times$	$\approx K = 8.0\times$	Calibrated ($\tau_{\text{iso}} = 2.0$)
GEPA (§4.2)	Max angular deviation ($\sigma = 0.5, \theta_{\max} = 5.0°$, 64 trials)	$5.0000°$	$\leq 5.0000°$	100% In-Bounds
	Child unit-sphere norm $\|\mathbf{c}\|_2$ (64 trials)	1.000000	1.0 ± 10^{-6}	100% Normalized
	Anchor penalty $\beta(1 - \cos(\mathbf{x}, \mathbf{x}_{\text{seed}}))$ ($\beta = 0.5, \theta = 90°$)	0.5000	0.5000	Deterministic

$0.4226K$. *In particular, for any* $K \geq 8$, $\mathbb{E}[I_{EP}(\mathbf{X}_{coll})] \geq 3.381 > \tau_{iso} = 2.0$, *whereas for an i.i.d. standard normal slice* $z_i \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$, $\mathbb{E}[T_{EP}(\mathbf{z})] = 1 - \frac{1}{\sqrt{3}} \approx 0.4226$ *independent of* K.

Proof. When $z_1 = \cdots = z_K = z_m$, every pairwise difference $z_i - z_j = 0$, so the double sum in Eq. (7) equals $\frac{1}{K}\sum_{i,j} e^0 = K$. The cross term equals $K\sqrt{2}e^{-z_m^2/4}$, yielding $T_{\text{EP}}(\mathbf{z}^{(m)}) = K(1 - \sqrt{2}e^{-z_m^2/4} + 1/\sqrt{3})$. Since $e^{-z_m^2/4} \leq 1$, the minimum over all $z_m \in \mathbb{R}$ is $K(1 + 1/\sqrt{3} - \sqrt{2}) \approx 0.1631K$. Taking expectation over $z_m \sim \mathcal{N}(0,1)$ uses $\mathbb{E}[e^{-z^2/4}] = (1+2\cdot\frac{1}{4})^{-1/2} = \sqrt{2/3}$, giving $\mathbb{E}[T_{\text{EP}}] = K(1+1/\sqrt{3} - \sqrt{4/3}) \approx 0.4226K$. Conversely, for $z_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, the K diagonal terms ($i = j$) contribute 1, the $K(K-1)$ off-diagonal terms with $z_i - z_j \sim \mathcal{N}(0,2)$ have $\mathbb{E}[e^{-(z_i - z_j)^2/2}] = (1+2)^{-1/2} = 1/\sqrt{3}$, and the cross term has $\mathbb{E}[\sqrt{2}\sum_i e^{-z_i^2/4}] = K\sqrt{4/3} = \frac{2K}{\sqrt{3}}$, so $\mathbb{E}[T_{\text{EP}}] = 1 + \frac{K-1}{\sqrt{3}} - \frac{2K}{\sqrt{3}} + \frac{K}{\sqrt{3}} = 1 - \frac{1}{\sqrt{3}} \approx 0.4226$ for all K. □

Proposition 3 (Monotonic Non-Regression of the Fail-Closed Gate)**.** *Let* $c_0, c_1, \ldots, c_T$ *be a sequence of target-ref states produced by* T *invocations of* `run_gate` *with* $c_0 = c_{target}^{(0)}$. *For every step* t *where promotion occurs* ($c_t \neq c_{t-1}$), *every tracked metric* $p \in P_{\max} \cup P_{\min}$ *satisfies* $\Delta_p(c_{t-1}, c_t) \geq -\epsilon$, *at least one metric* p^* *satisfies* $\Delta_{p^*}(c_{t-1}, c_t) > \epsilon$, *and the population isotropy satisfies* $I_{EP}(\mathbf{X}_t) \leq \tau_{iso}$.

6 Experimental Evaluation

We evaluate LEIO-HARNESS ('v2.6.5') across three dimensions: (1) statistical discrimination of collapsed vs. isotropic multi-agent populations and trust-region compliance, (2) multi-model parallel synthesis sweeps across 88 lanes spanning 8 LLM families, and (3) coordination bandwidth and latency over the Arrow Flight bus.

6.1 SIGReg Collapse Detection and EmbeddingGEPA Verification

Table 1 reports the measured behavior of the live `leio-harness sigreg isotropy` and `leio-harness gepa` primitives on $K = 8$ concurrent agent lanes ($d = 8$, $M = 64$ projection slices, seed 42). Confirming Proposition 2, a directionally collapsed 8-lane population yields $I_{\text{EP}} = 3.5042$ (matching the theoretical asymptotic prediction $0.4226 \times 8 = 3.381$ within finite-d variance) and triggers the `Collapsed` gate verdict ($I_{\text{EP}} > 2.0$). Conversely, an isotropically dispersed 8-lane population yields $I_{\text{EP}} = 0.3432 \leq 2.0$, passing the gate with a $10.2\times$ separation ratio. Across 64 consecutive tangent-space mutations with $\theta_{\max} = 5.0°$ and $\sigma = 0.5$ ($\arctan(0.5) = 26.57°$ unconstrained), `trust_region_mutate` clips 100% of children to $5.000° \pm 10^{-4°}$ with unit norm $\|\mathbf{c}\|_2 = 1.000000$.

Table 2: 48-lane parallel multi-model sweep (8 models × 6 tasks: 5 LEIO repository golden tasks + 1 cross-source governance task) executed via `leio-harness day`.

Model Family (Lane Worker)	JSON Valid	Code Evidence	Gov. Evidence	Tests Hit	Mean Wall Clock
anthropic/claude-fable-5.1	6/6 (100.0%)	0.197	**0.571**	5/5 (100%)	12.4 s
openai/gpt-5.6-luna	6/6 (100.0%)	**0.231**	0.429	5/5 (100%)	9.8 s
anthropic/claude-opus-5	6/6 (100.0%)	0.214	0.429	5/5 (100%)	14.1 s
x-ai/grok-4.3-surge	6/6 (100.0%)	0.188	0.286	5/5 (100%)	8.9 s
z-ai/glm-5.3	6/6 (100.0%)	0.179	0.286	5/5 (100%)	7.6 s
moonshotai/kimi-k2.7	6/6 (100.0%)	0.165	0.286	4/5 (80%)	8.2 s
deepseek/deepseek-v4-flash-0731	6/6 (100.0%)	0.160	0.286	5/5 (100%)	4.3 s
qwen/qwen3.8-flash (baseline)	4/5 (80.0%)	0.089	0.143	3/5 (60%)	2.9 s
qwen/qwen3.8-flash + category-scan + json_schema	**6/6 (100.0%)**	**0.261**	**0.571**	**5/5 (100%)**	**3.1 s**

Table 3: Prompt treatment and structured decoding ablation on `qwen/qwen3.8-flash` across the 6-task suite.

Prompt Treatment	Response Format	JSON Valid	Code Evidence Ratio	Gov. Evidence Ratio
baseline	none	4/5 (80.0%)	0.089	0.143
schema-last	json_schema	6/6 (100.0%)	0.142 (+59.6%)	0.286
citation-checklist	json_schema	6/6 (100.0%)	0.218 (+144.9%)	0.429
schema-last+checklist	json_schema	6/6 (100.0%)	0.229 (+157.3%)	0.429
category-scan (Default)	**json_schema**	**6/6 (100.0%)**	**0.261 (+193.3%)**	**0.571 (+299.3%)**

6.2 Multi-Model Parallel Synthesis Sweep (88 Lanes Across 8 Models)

Using `scripts/benchmark_models.py` (`leio-bench`), we execute parallel synthesis sweeps through `leio-harness day` across 8 diverse frontier and flash models routed via OpenRouter: `claude-fable-5.1`, `claude-opus-5`, `gpt-5.6-luna`, `grok-4.3-surge`, `glm-5.3`, `kimi-k2.7`, `deepseek-v4-flash-0731`, and `qwen3.8-flash`. Each lane receives an isolated worktree, queries the structured LEIO Code context bundle ('leio-code context –json'), and emits a strict JSON synthesis verdict `{"answer", "evidence_paths", "tests"}`. We grade four metrics per lane: schema compliance (`json_valid`), grounded file citation recall (`evidence_ratio` on repository code tasks and cross-source reference tasks), verification command accuracy (`tests_hit`), and wall-clock latency (`duration_ms`).

Closing the Small-Model Grounding Gap via Structured Zone Scanning. As shown in Table 2, under the baseline prompt treatment, the fastest flash model (`qwen3.8-flash`, 2.9 s/lane) lagged frontier models in both JSON compliance (80%) and code evidence recall (0.089 vs. 0.197 for `claude-fable-5.1`). By systematically ablating five prompt and decoding treatments across 30 additional lanes (Table 3), we found that combining strict grammar-constrained decoding (`response_format=json_schema`) with zone-by-zone context bundle scanning (`category-scan`: explicitly scanning instructions, memory, anchors, ranked files, verification, and risks) eliminates format failures (6/6 JSON valid) and boosts `qwen3.8-flash` code evidence recall by +193% (0.089 → 0.261) and governance recall by +299% (0.143 → 0.571), outperforming the baseline frontier models at 4× lower latency (3.1 s vs. 12.4 s).

6.3 Coordination Bandwidth and Latency Analysis

In a conventional $K = 8$ conversational multi-agent topology where each agent broadcasts a 1,500-token status/diff summary across $R = 3$ coordination rounds, peer-to-peer transcript sharing consumes $R \cdot K(K-1)\bar{L} = 3 \times 56 \times 1{,}500 = 252{,}000$ LLM context tokens per run. In LEIO-HARNESS, inter-agent coordination over the Arrow Flight bus transmits zero LLM tokens (0 conversational

tokens) and only $8 \times (64 + 4d)$ bytes of Arrow `RecordBatch` binary payload (1.5 KiB total for $d = 32$ or 12.5 KiB for $d = 384$), completing `do_put`, `do_get`, and `do_exchange` cosine top-k matching in < 1.8 ms on local loopback.

7 Conclusion

We presented LEIO-HARNESS, a multi-agent code synthesis runtime that replaces quadratic natural-language chatter with zero-copy Apache Arrow Flight state embeddings, geodesic trust-region evolution (EMBEDDINGGEPA), sliced Epps–Pulley isotropy collapse detection (SIGREG), and a fail-closed Pareto integration gate. Both theoretical analysis and 88-lane multi-model empirical sweeps demonstrate that geometric coordination and fail-closed verification enable fast, diverse, and provably non-regressing multi-agent software engineering.

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